Statistical models

Exercise Sheet 3

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Solve the following problems before the exercise class on May 20.

Problem 1
For $\beta, \mathbf{x} \in \mathbb{R}^k$ with $k \in \mathbb{N}$ and a random variable ε with distribution function F let

$$Z = \beta^T \mathbf{x} + \varepsilon \quad \text{and} \quad Y = \begin{cases} 1, & Z > 0, \\ 0, & Z \leq 0. \end{cases}$$

Compute for

a) $F(x) = \frac{1}{1+e^{-x}}$ for $x \in \mathbb{R}$

b) $F = \Phi$, where Φ is the distribution function of the standard normal distribution,

the probability $\mathbf{P}(Y = 1)$.

c) Which interpretations for logistic and probit regression do we obtain from parts a) and b)?

Problem 2
The data set `babyfood` from the `R` package `faraway` provides the numbers of children developping a respiratory disease in their first year of life depending on the sex and the type of feeding. The factor `sex` has the levels `Boy` and `Girl`, while the factor `food` has the levels `Bottle`, `Breast` and `Suppl` (for breast feeding with supplement).
In the following, we apply logistic regression to study the probability of a child having the disease. Use `R` for the computations and test all null hypotheses at level 0.01 (without correcting for multiple testing).

a) Formulate a model for logistic regression (without interaction).

b) Estimate the parameters.

c) Test the null hypotheses that the parameters are zero.

d) Provide the probability that a bottle-fed girl gets the disease.

e) Formulate a model for logistic regression with interaction and estimate the parameters. What are here the fitted values?

Problem 3

Recall that a random variable Z follows a Poisson distribution with parameter $\lambda \in (0, \infty)$ if

$$\mathbf{P}(Z = k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

for all $k \in \mathbb{N}_0 := \mathbb{N} \cup \{0\}$. We use the short hand notation $Z \sim \text{Poisson}(\lambda)$. In Poisson regression, the dependent variable Y is assumed to satisfy $Y \sim \text{Poisson}(\exp(\beta^T \mathbf{x}))$ for some $\beta \in \mathbb{R}^k$, where $\mathbf{x} \in \mathbb{R}^k$ is the vector of the independent variables $x^{(1)}, \ldots, x^{(k)}$. Let observations $(Y_1, \mathbf{x}_1), \ldots, (Y_n, \mathbf{x}_n)$ of $(Y, \mathbf{x})$ be given such that $Y_1, \ldots, Y_n$ are independent. Compute the maximum likelihood estimator $\widehat{\beta}$ for β.